基于多模态大模型与 Skill 的肝脏超声包虫病识别探索

1 任务与核心问题

本项目完成了以下工作：

- 构建包含 166 张图像的 liver ultrasound benchmark；
- 建立统一的 multimodal LLM Agent pipeline；
- 比较 GLM-5.3-Flash 与 DeepSeek-V4-Flash-Vision-Exp；
- 设计并迭代 Skill v1–v4；
- 冻结实验方案后，在 held-out test 上完成最终评估。

Skill 是附加在任务 prompt 中的文本判读流程，不是训练、微调或新的分类器。本项目的核心问题是：**额外 Skill 对不同基础多模态模型的影响是否一致？**

2 数据与实验设计

阳性数据。包虫病图像来自 Bioengineering (2025) 论文《A Study on Staging Cystic Echinococcosis Using Machine Learning Methods》(DOI: 10.3390/bioengineering12020181；论文链接，Supplementary dataset)。本项目使用其 Supplementary Material 1 中的超声数据；本地的 `Dataset2_US.pdf` 即来自该补充材料。从补充 PDF 提取图像后，排除确认不是肝脏超声的 CT/X-ray，并在去重后得到 83 张包虫病超声图像。

阴性数据。非包虫病图像来自 Zenodo 数据集《Annotated Ultrasound Liver images》（DOI: 10.5281/zenodo.7272660；Zenodo 数据集）。该数据集包含 Normal、Benign 和 Malignant 三类；本项目从中固定抽取 83 张，作为非包虫病肝脏超声。其对应论文为《Improving artificial intelligence pipeline for liver malignancy diagnosis using ultrasound images and video frames》（DOI: 10.1093/bib/bbac569；对应论文）。

因此，本文的 “阴性” 并不等于健康人：其中既有正常超声，也有良性和恶性肝脏病变。阳性与阴性来自不同公开来源，**数据来源差异（dataset-source / domain bias）**是当前 benchmark 最重要的局限之一。

Split	Positive	Negative	Total
dev	50	50	100
held-out test	33	33	66

dev 用于 Skill 设计迭代，held-out test 在开发过程中保持隔离。两种模型使用相同的任务说明、数据划分、结构化输出和评估方式。实验流程为：

`Image + task prompt + optional Skill` → `Multimodal LLM` → `structured prediction` → `evaluation`

3 Baseline 的不同错误画像

在介绍 Skill 前，先比较两个模型在当前 held-out benchmark 上观察到的 baseline error pattern。本文将包虫病图像定义为阳性（positive），非包虫病肝脏超声定义为阴性（negative）。FN（假阴性）表示阳性图像被误判为阴性；FP（假阳性）表示阴性图像被误判为阳性。

Model	准确率	灵敏度	特异度	FN（假阴性）	FP（假阳性）
GLM baseline	93.94%	87.88%	100.00%	4	0
DeepSeek baseline	71.21%	96.97%	45.45%	1	18

在当前 test 上，GLM baseline 的 FP 为 0，主要剩余错误是 FN；DeepSeek baseline 的 FN 很少，但存在大量 FP，即明显的阳性过判。两个基础模型在加入 Skill 前面对的主要问题已经不同。这里描述的是当前 benchmark 上的观察结果，不代表模型在其他数据上的固定特征。

4 Skill v1–v4 的设计迭代

Skill 的设计迭代逻辑是：根据上一版本在 dev 上观察到的问题，设计下一版本进行检验；当前实验不据此建立严格因果关系。

专业版：设计维度对比 / Professional design-dimension comparison

比较内容	v1	v2	v3	v4
是否提供包虫病典型超声特征清单	是，清单较详细	是，保留 v1 清单	否，不再列出疾病特征清单	否，保持 v3 做法
图像不清楚时是否容易拒绝判断	相对容易：需看清肝脏或疑似病灶	不因低对比、非典型或病灶不明显而直接拒绝	只有明显不是肝脏超声，或完全无法判断时才拒绝	与 v3 相同
看不到某个典型特征时能否直接判阴性	没有专门限制	不能仅凭一个特征看不到就排除	不能仅凭一个特征看不到就排除	与 v3 相同
是否要求先描述图像，再作判断	是	是	是	是
是否同时防止“误报”和“漏报”	主要强调防止误报	仍主要强调防止误报，并减少过严排除	是，明确同时避免误报和漏报	是，保持 v3 做法
是否提供一个完整判断示例	否	否	否	是，示范从观察到结论的过程
是否提醒模型不要机械照搬示例	否	否	否	是

表 1: 四个 Skill 版本的设计维度对比。

开发集观察与下一步设计

Version	核心设计	观察与下一步
v1	显式 CE morphology checklist，并强调避免过度诊断	GLM sensitivity 明显下降，表现过于保守；下一步放宽强排除条件
v2	放宽 checklist 中的强排除条件，减少不必要拒判	sensitivity 仍未恢复，说明问题不只是个别措辞；下一步移除长疾病 checklist
v3	仅保留 systematic observation 与 evidence integration	GLM dev 恢复到 baseline 附近；DeepSeek FP 明显减少；下一步检验示例化 evidence integration
v4	v3 加入 worked reasoning example	检验示例是否进一步有效；最终效果仍表现出 model dependence

表 2: Skill v1–v4 的设计迭代逻辑。

GLM 在同一 dev100 上的严格准确率为：baseline 85%、v1 78%、v2 77%、v3 85%。这组结果解释了为什么从显式 checklist 转向更精简的结构化观察流程。DeepSeek dev100 的 FP 在 baseline 与 v3 间由 34 降至 24。

v4 的 worked example 来源于 dev 中一张图像的 source-provided morphology description，因此该样本未计入 v4 的开发集评估；v4 使用 99 张的公平评估集合，详细说明见附录。

5 Held-out test 最终结果

最终评估使用固定的 66 张 held-out test（33 positive、33 negative）。六个条件覆盖相同图像，均无 `not_applicable` 或解析失败。

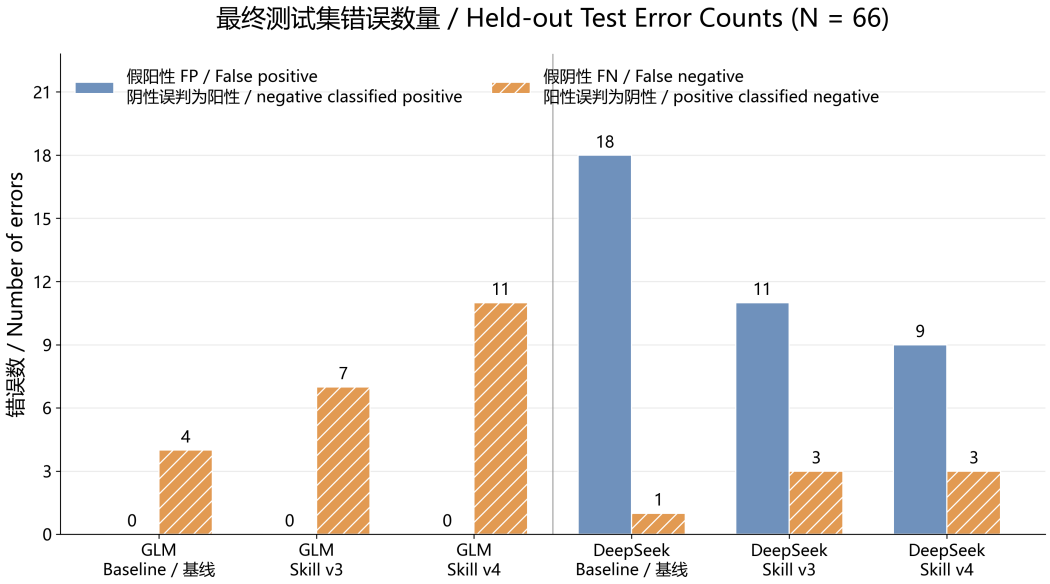


图 1: Held-out test 错误数量。FP（假阳性）表示阴性被误判为阳性；FN（假阴性）表示阳性被误判为阴性。

模型	条件	TP 正确识别包虫病	FN 漏报	FP 误报	TN 正确识别非包虫病	准确率	灵敏度	特异度
GLM	baseline	29	4	0	33	93.94%	87.88%	100.00%
GLM	Skill v3	26	7	0	33	89.39%	78.79%	100.00%
GLM	Skill v4	22	11	0	33	83.33%	66.67%	100.00%
DeepSeek	baseline	32	1	18	15	71.21%	96.97%	45.45%
DeepSeek	Skill v3	30	3	11	22	78.79%	90.91%	66.67%
DeepSeek	Skill v4	30	3	9	24	81.82%	90.91%	72.73%

表 3: 六个条件在 held-out test 上的主要指标。

GLM： FN 从 4 增至 7、11，FP 始终为 0。在当前 test 上，GLM baseline 已经没有 FP；v3/v4 没有带来额外 specificity 收益，主要增加 FN，使 sensitivity 和 accuracy 下降。

DeepSeek： FP 从 18 降至 11、9，FN 从 1 增至 3 后保持不变。v3/v4 主要减少 FP，同时带来少量 FN 增加，因此 specificity 与 overall accuracy 上升，sensitivity 略有下降。

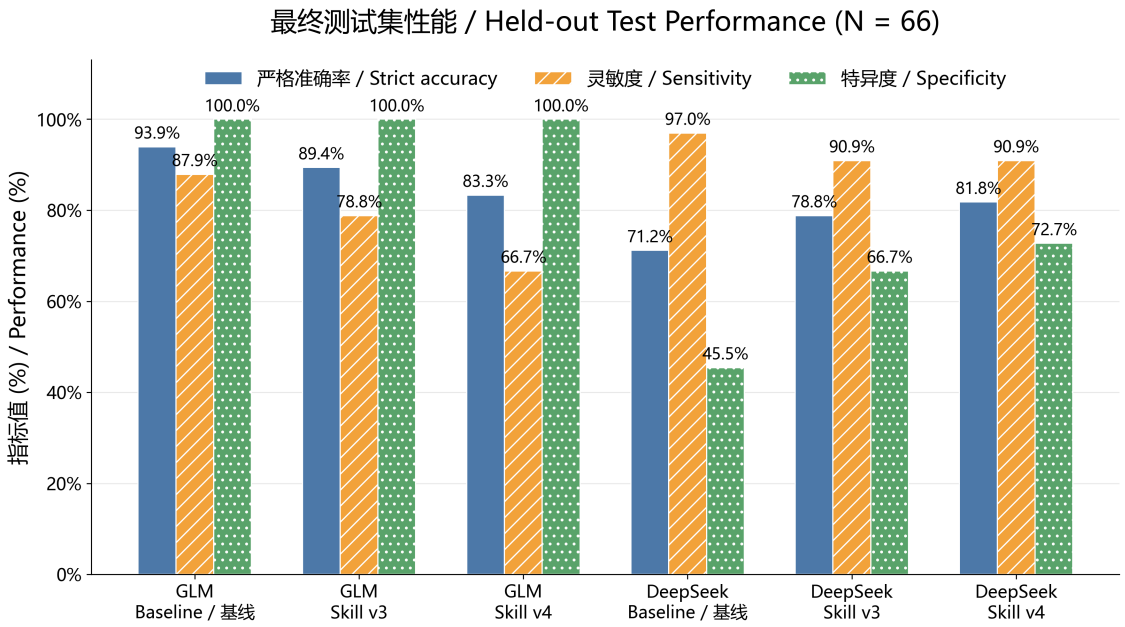


图 2: 六个 held-out test 条件的严格准确率（Strict accuracy）、灵敏度（Sensitivity）与特异度（Specificity）。

图 2 从比例指标上补充显示：GLM 的主要变化是 sensitivity 下降；DeepSeek 的主要变化是 specificity 上升。

6 核心发现、局限与下一步

1. **Skill 不是普遍增益**。在当前 benchmark 上，GLM baseline 优于其 Skill 条件，而 DeepSeek 的 observed accuracy 在 v3/v4 下提高。
2. **Skill 更像改变模型的 observed decision behavior**。主要表现为 sensitivity–specificity trade-off，而不能简单解释为“增加了医学知识”。
3. **Skill 效果依赖基础模型原本的 error pattern**。两个 baseline 面对的主要错误不同，阳性预测倾向下降因而产生不同后果。

Comparison	FP	FN	Accuracy
GLM baseline → v4	$0 \to 0$	$4 \to 11$	$93.94\% \to 83.33\%$
DeepSeek baseline → v4	$18 \to 9$	$1 \to 3$	$71.21\% \to 81.82\%$

因此，同样表现为“阳性预测倾向下降”的提示变化，在 DeepSeek 上主要修正 false positives，但在 GLM 上主要增加 false negatives。worked reasoning example 也没有表现出跨模型一致收益：相对 v3，v4 在 GLM 上降低 accuracy，在 DeepSeek 上提高 accuracy。

DeepSeek baseline vs v4 的 McNemar exact $p = 0.092285$，因此当前结果不能表述为 statistically significant improvement。

6.1 主要局限

- 数据规模较小，且只使用一个固定的开发集/最终测试集划分；

- 阳性与阴性来自不同公开来源，存在数据来源差异；
- 输入为单张静态超声图像，缺少完整临床信息和独立医生复核。

因此，本实验是探索性小型 benchmark，不能用于推断真实临床诊断性能。

6.2 下一步讨论

1. 是否需要构建来源匹配的阳性/阴性数据，或设置独立外部测试集?
2. 是否应扩大到多中心、多设备、更多图像，甚至纳入多帧或动态超声?
3. 下一步 Skill 是否仍聚焦文字判读流程，还是转向多阶段 Agent / 工具辅助流程?

附录 / 补充结果

A. v4 的 dev99 公平评估说明

v4 的完整判断示例由开发集中一张图像的来源描述抽象改写而来。为避免该信息影响开发集评估，该图像未计入 v4 评估，形成 dev99（49 张包虫病、50 张非包虫病）；原始 dev100 和最终测试集均未修改。

模型	baseline	Skill v3	Skill v4
GLM	85.86%	85.86%	82.83%
DeepSeek	62.63%	69.70%	71.72%

B. 完整 held-out test 指标

Model	Condition	N	TP	FN	FP	TN	NA	PF	Accuracy	Precision	Sensitivity	Specificity	F1
GLM	baseline	66	29	4	0	33	0	0	93.94%	100.00%	87.88%	100.00%	93.55%
GLM	Skill v3	66	26	7	0	33	0	0	89.39%	100.00%	78.79%	100.00%	88.14%
GLM	Skill v4	66	22	11	0	33	0	0	83.33%	100.00%	66.67%	100.00%	80.00%
DeepSeek	baseline	66	32	1	18	15	0	0	71.21%	64.00%	96.97%	45.45%	77.11%
DeepSeek	Skill v3	66	30	3	11	22	0	0	78.79%	73.17%	90.91%	66.67%	81.08%
DeepSeek	Skill v4	66	30	3	9	24	0	0	81.82%	76.92%	90.91%	72.73%	83.33%

C. McNemar exact test

Model	Comparison	correct→wrong	wrong→correct	Exact two-sided p
GLM	baseline vs v3	6	3	0.507812
GLM	baseline vs v4	8	1	0.039062
GLM	v3 vs v4	7	3	0.343750
DeepSeek	baseline vs v3	3	8	0.226562
DeepSeek	baseline vs v4	3	10	0.092285
DeepSeek	v3 vs v4	4	6	0.753906

D. 三类阴性来源的分类正确率

33 张最终测试阴性样本包括 Normal（正常）5 张、Benign（良性病变）9 张和 Malignant（恶性病变）19 张。下表的“分类正确率”指每一类阴性样本中被正确判断为阴性的比例。

Model	Condition	Normal（$n = 5$）	Benign（$n = 9$）	Malignant（$n = 19$）
GLM	baseline	5/5（100.00%）	9/9（100.00%）	19/19（100.00%）
GLM	Skill v3	5/5（100.00%）	9/9（100.00%）	19/19（100.00%）
GLM	Skill v4	5/5（100.00%）	9/9（100.00%）	19/19（100.00%）
DeepSeek	baseline	3/5（60.00%）	5/9（55.56%）	7/19（36.84%）
DeepSeek	Skill v3	4/5（80.00%）	4/9（44.44%）	14/19（73.68%）
DeepSeek	Skill v4	4/5（80.00%）	6/9（66.67%）	14/19（73.68%）

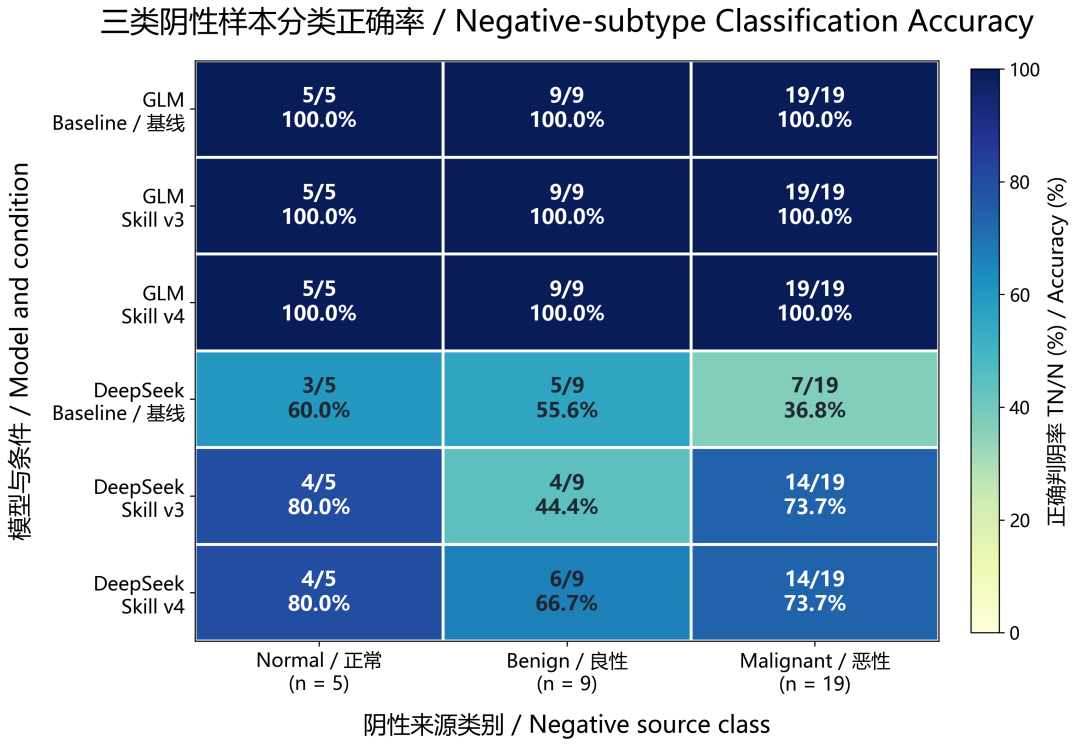


图 3: 六种模型/Skill 条件在三类阴性来源上的分类正确率。

E. Skill v1–v4 完整原文

以下内容按冻结的 `SKILL.md` 原文呈现，并保留元数据头。它们是在统一任务说明之外附加给模型的完整 Skill 文本；baseline 不包含这些附加内容。

Skill v1

```
---
name: echo-hydatid-v1
description: Apply a structured liver-ultrasound review workflow for experimental
↪  echinococcosis-versus-negative classification while preserving the existing output schema.
```

```
---

# Echo Hydatid Skill v1

## Goal

- Review one liver ultrasound image using a consistent, observation-first workflow.
- Make the primary binary decision: `echinococcosis` or `negative`.
- Use this skill only as an additional domain-specific assessment procedure. It does not replace the baseline
↪  task instructions, create a new classifier, or change the existing JSON output schema.
- Treat WHO stage as optional supporting information. Do not require a stage to make the binary decision.

## Step 1: Validate the image

Before interpreting pathology, confirm both of the following:

- The input is an interpretable liver ultrasound image.
- Liver parenchyma or a suspected focal lesion is sufficiently visible for assessment.

If the image is not an analyzable liver ultrasound, stop the medical assessment and return:

- `image_valid = false`
- `diagnosis = not_applicable`
- `subtype = null`
- `confidence = null`

Do not continue with diagnostic inference after declaring the image invalid.

## Step 2: Observe before diagnosing

Follow the rule: **observation first, diagnosis second**. Record only features that are visible in the
↪  supplied image. Review the image in this order:

1. Determine whether a focal lesion is present.
2. If present, describe it as `cystic`, `solid`, or `mixed`.
3. Assess whether the lesion is `unilocular` or `multivesicular`.
4. Describe the cyst wall, lesion boundary, or both.
5. Look for internal membranes.
6. Look for daughter cysts or daughter vesicles.
7. Assess internal echoes or heterogeneous material.
8. Look for calcification.
9. Look for posterior acoustic shadowing.

Do not infer features that are outside the field of view. Do not treat a possible observation as confirmed
↪  merely because it would fit a diagnosis.

## Step 3: Look for characteristic CE morphology

When assessing possible cystic echinococcosis (CE), give particular attention to the following morphology:

- Double-wall morphology.
- Hydatid sand or characteristic fine internal echoes.
- Daughter cysts.
- A multivesicular honeycomb or rosette-like appearance.
- A detached internal membrane or water-lily morphology.
- Daughter vesicles within a predominantly solid matrix.
- Characteristic degenerative ball-of-wool or canalicular morphology.
- A near-complete eggshell-calcified wall with posterior acoustic shadowing.
```

```
These findings increase support for CE, but no item in this list should automatically determine the diagnosis
↪  by itself. Integrate the complete visible morphology, the consistency of the findings, and the
↪  limitations of the single image.

## Step 4: Avoid overdiagnosis

A simple round or oval anechoic cyst, without characteristic CE morphology, is not sufficient by itself to
↪  diagnose echinococcosis.

Calcification alone is not sufficient evidence to determine CE diagnosis, stage, or viability.

Do not convert a nonspecific focal liver finding into an echinococcosis diagnosis solely because it is
↪  cystic. Distinguish characteristic CE morphology from findings that may occur in other liver lesions.

## Step 5: Integrate the evidence

Make the final binary decision from the overall pattern rather than from a single isolated feature:

- If characteristic CE morphology is clearly present, use `diagnosis = echinococcosis`.
- If multiple mutually consistent CE features are present, use `diagnosis = echinococcosis`.
- If only nonspecific cystic or focal liver findings are present and characteristic CE morphology is absent,
↪  do not infer echinococcosis from cystic morphology alone.
- Otherwise, use `diagnosis = negative`.

Keep this integration qualitative. Do not create a numeric score, assign feature weights, or invent decision
↪  thresholds. Express uncertainty in the existing `confidence` output, but do not describe self-reported
↪  confidence as a calibrated probability.

## Step 6: WHO stage

WHO stage is optional. Assign `CE1`, `CE2`, `CE3a`, `CE3b`, `CE4`, or `CE5` only when the visible morphology
↪  clearly supports that stage.

If the stage is not sufficiently clear, use `subtype = null`. Do not force a stage merely to produce a more
↪  specific answer.

A single static ultrasound image cannot reliably establish signs that require dynamic observation. Do not
↪  infer movement or other dynamic behavior from a static frame.

## Output constraints

- Continue to use the existing JSON schema and only these fields: `image_valid`, `diagnosis`, `confidence`,
↪  `subtype`, and `findings`.
- Do not add new output fields.
- `diagnosis` must be exactly one of: `echinococcosis`, `negative`, or `not_applicable`.
- When the image is invalid, use the values specified in Step 1.
- Use `subtype = null` whenever a WHO stage is not clearly supported.
- Keep `findings` focused on visible observations and make the diagnosis only after those observations have
↪  been integrated.
- Treat `confidence` as the model's self-reported certainty, not as a calibrated probability.

## References / provenance note

This skill is intended to summarize a structured ultrasound assessment workflow derived from established CE
↪  ultrasound morphology and WHO-IWGE style classification concepts. It should be treated as an experimental
↪  decision-support prompt, not as a clinical diagnostic guideline.
```

Skill v2

```
---
name: echo-hydatid-v2
description: Apply a structured liver-ultrasound review workflow for experimental
↪  echinococcosis-versus-negative classification while preserving the existing output schema.
---

# Echo Hydatid Skill v2

## Goal

- Review one liver ultrasound image using a consistent, observation-first workflow.
- Make the primary binary decision: `echinococcosis` or `negative`.
- Use this skill only as an additional domain-specific assessment procedure. It does not replace the baseline
↪  task instructions, create a new classifier, or change the existing JSON output schema.
- Treat WHO stage as optional supporting information. Do not require a stage to make the binary decision.

## Step 1: Validate the image

Before interpreting pathology, confirm both of the following:

- The input can reasonably be interpreted as a liver ultrasound image.
- The image is technically interpretable enough to assess the visible morphology.

Use `not_applicable` only when the image is clearly not a liver ultrasound or is technically uninterpretable.

Low contrast, atypical appearance, or difficulty identifying a lesion alone is not sufficient reason to mark
↪  the image invalid.

When `not_applicable` is appropriate, stop the medical assessment and return:

- `image_valid = false`
- `diagnosis = not_applicable`
- `subtype = null`
- `confidence = null`

Do not continue with diagnostic inference after declaring the image invalid.

## Step 2: Observe before diagnosing

Follow the rule: **observation first, diagnosis second**. Record only features that are visible in the
↪  supplied image. Review the image in this order:

1. Determine whether a focal lesion is present.
2. If present, describe it as `cystic`, `solid`, or `mixed`.
3. Assess whether the lesion is `unilocular` or `multivesicular`.
4. Describe the cyst wall, lesion boundary, or both.
5. Look for internal membranes.
6. Look for daughter cysts or daughter vesicles.
7. Assess internal echoes or heterogeneous material.
8. Look for calcification.
9. Look for posterior acoustic shadowing.

Do not infer features that are outside the field of view. Do not treat a possible observation as confirmed
↪  merely because it would fit a diagnosis.

## Step 3: Look for characteristic CE morphology
```

```
When assessing possible cystic echinococcosis (CE), give particular attention to the following morphology:

- Double-wall morphology.
- Hydatid sand or characteristic fine internal echoes.
- Daughter cysts.
- A multivesicular honeycomb or rosette-like appearance.
- A detached internal membrane or water-lily morphology.
- Daughter vesicles within a predominantly solid matrix.
- Characteristic degenerative ball-of-wool or canalicular morphology.
- A near-complete eggshell-calcified wall with posterior acoustic shadowing.

These findings increase support for CE, but no item in this list should automatically determine the diagnosis
↪  by itself. Integrate the complete visible morphology, the consistency of the findings, and the
↪  limitations of the single image.

## Step 4: Avoid overdiagnosis

A simple round or oval anechoic cyst, without characteristic CE morphology, is not sufficient by itself to
↪  diagnose echinococcosis.

Calcification alone is not sufficient evidence to determine CE diagnosis, stage, or viability.

Do not convert a nonspecific focal liver finding into an echinococcosis diagnosis solely because it is
↪  cystic. Distinguish characteristic CE morphology from findings that may occur in other liver lesions.

## Step 5: Integrate the evidence

Make the final binary decision from the overall pattern rather than from a single isolated feature:

- If characteristic CE morphology is clearly present, use `diagnosis = echinococcosis`.
- If multiple mutually consistent CE features are present, use `diagnosis = echinococcosis`.
- If the visible evidence consists only of nonspecific cystic or focal liver findings, do not infer
↪  echinococcosis from cystic morphology alone.
- After integrating all visible positive and nonspecific evidence, if the overall morphology does not support
↪  CE, use `diagnosis = negative`.

The absence or non-visibility of any single characteristic feature should not by itself be treated as
↪  evidence against echinococcosis.

Do not require all or most listed CE features to be present.

Judge the image from the overall visible morphology and any positive supporting evidence that is actually
↪  present.

Keep this integration qualitative. Do not create a numeric score, assign feature weights, or invent decision
↪  thresholds. Express uncertainty in the existing `confidence` output, but do not describe self-reported
↪  confidence as a calibrated probability.

## Step 6: WHO stage

WHO stage is optional. Assign `CE1`, `CE2`, `CE3a`, `CE3b`, `CE4`, or `CE5` only when the visible morphology
↪  clearly supports that stage.

If the stage is not sufficiently clear, use `subtype = null`. Do not force a stage merely to produce a more
↪  specific answer.
```

```
A single static ultrasound image cannot reliably establish signs that require dynamic observation. Do not
↪  infer movement or other dynamic behavior from a static frame.

## Output constraints

- Continue to use the existing JSON schema and only these fields: `image_valid`, `diagnosis`, `confidence`,
↪  `subtype`, and `findings`.
- Do not add new output fields.
- `diagnosis` must be exactly one of: `echinococcosis`, `negative`, or `not_applicable`.
- When the image is invalid, use the values specified in Step 1.
- Use `subtype = null` whenever a WHO stage is not clearly supported.
- Keep `findings` focused on visible observations and make the diagnosis only after those observations have
↪  been integrated.
- Treat `confidence` as the model's self-reported certainty, not as a calibrated probability.

## References / provenance note

This skill is intended to summarize a structured ultrasound assessment workflow derived from established CE
↪  ultrasound morphology and WHO-IWGE style classification concepts. It should be treated as an experimental
↪  decision-support prompt, not as a clinical diagnostic guideline.
```

Skill v3

```
---
name: echo-hydatid-v3
description: Apply a minimal structured-observation workflow for experimental liver-ultrasound classification
↪  without an explicit disease-feature checklist.
---

# Echo Hydatid Skill v3

## Goal

- Review one liver ultrasound image in a consistent, observation-first order.
- Make the primary binary decision: `echinococcosis` or `negative`.
- Use this skill to structure image review, not to replace the baseline task or teach a complete body of
↪  medical knowledge.
- Do not force a subtype when the image does not clearly support one.

## Step 1: Validate the image

Use `diagnosis = not_applicable` only when:

- The image is clearly not a liver ultrasound; or
- Technical quality is so poor that no meaningful visual assessment is possible.

Do not mark the image invalid merely because contrast is low, the appearance is atypical, a lesion is subtle,
↪  or the interpretation is uncertain. If a meaningful assessment remains possible, continue with the binary
↪  task.

## Step 2: Observe systematically

Observe before deciding. Describe only what is visible:

1. Whether a focal lesion is present.
2. Whether the lesion appears `cystic`, `solid`, or `mixed`.
3. The lesion boundary or wall.
4. Its internal architecture.
```

```
5. Any visible internal echoes or structures.
6. The appearance of the surrounding liver.

Keep observations separate from the final diagnosis. Do not infer findings outside the supplied frame, and do
↪  not turn this sequence into a disease-feature checklist.

## Step 3: Integrate evidence

Base the diagnosis on the overall visible morphology. Use positive evidence that is actually present.

Do not treat the absence of one expected feature as sufficient evidence against echinococcosis. Do not
↪  require a fixed number of characteristic findings.

Do not create a score, assign feature weights, or invent a decision threshold. Integrate the visible evidence
↪  qualitatively, then choose the diagnosis that is best supported by the complete image.

## Step 4: Avoid two opposite errors

- **Error A — overdiagnosis:** Do not diagnose echinococcosis from a generic or simple cystic lesion alone.
- **Error B — underdiagnosis:** Do not reject echinococcosis merely because one or more classic-looking
↪  features are not visible in a single static image.

A single static ultrasound image may not show every characteristic feature. Balance positive and negative
↪  evidence before making the final decision.

## Output constraints

- Use only the existing fields: `image_valid`, `diagnosis`, `confidence`, `subtype`, and `findings`.
- Do not add fields.
- `diagnosis` must be exactly `echinococcosis`, `negative`, or `not_applicable`.
- When the image is invalid, use `image_valid = false`, `diagnosis = not_applicable`, `subtype = null`, and
↪  `confidence = null`.
- Set `subtype = null` unless a subtype is very clearly supported.
- Keep `findings` concise and grounded in visible observations.
- Treat `confidence` as self-reported confidence, not as a calibrated probability.

## Provenance note

This skill is an experimental structured-observation prompt. It intentionally avoids an explicit
↪  disease-feature checklist in order to test whether reasoning structure alone improves performance.
```

Skill v4

```
---
name: echo-hydatid-v4
description: Apply structured liver-ultrasound observation with a generalized evidence-integration example
↪  and an explicit bias counterbalance.
---

# Echo Hydatid Skill v4

## Goal

- Review one liver ultrasound image in an observation-first order.
- Make the primary binary decision: `echinococcosis` or `negative`.
- Structure observation and evidence integration without replacing the baseline task or teaching a complete
↪  body of medical knowledge.
- Do not force a subtype when the image does not clearly support one.
```

```
## Step 1: Validate the image

Use `diagnosis = not_applicable` only when:

- The image is clearly not a liver ultrasound; or
- Technical quality is so poor that no meaningful visual assessment is possible.

Low contrast, atypical appearance, a subtle lesion, or uncertainty alone does not make the image invalid.
↪  Continue whenever meaningful assessment remains possible.

## Step 2: Observe systematically

Observe before deciding. Describe only what is visible:

1. Whether a focal lesion is present.
2. Whether it appears `cystic`, `solid`, or `mixed`.
3. The lesion boundary or wall.
4. Its internal architecture.
5. The appearance of the surrounding liver.

Keep observation separate from diagnosis. Do not infer beyond the supplied frame or turn this sequence into a
↪  disease-feature checklist.

## Step 3: Integrate evidence

Base the diagnosis on the overall visible morphology and evidence actually present.

Ask whether observations form a coherent pattern, considering supporting and conflicting evidence. One absent
↪  expected feature is not sufficient evidence against echinococcosis. Do not require a fixed number of
↪  findings.

Do not create a score, feature weights, or a threshold. Integrate evidence qualitatively and choose the
↪  best-supported diagnosis.

## Step 4: Avoid two opposite errors

- **Overdiagnosis:** Do not diagnose echinococcosis from a generic or simple cystic lesion alone.
- **Underdiagnosis:** Do not reject echinococcosis merely because one or more expected-looking features are
↪  not visible in a single static image.

Balance positive, negative, and uncertain observations before deciding from a static image.

## Worked reasoning example

This demonstrates evidence integration, not a reusable disease rule.

**Observation**

The image is an interpretable liver ultrasound. A focal cystic lesion is visible. It is not internally
↪  uniform: several organized components are present within heterogeneous material. Its boundary has a
↪  layered appearance. Record these observations before assigning a diagnosis.

**Evidence integration**
```

```
Do not decide from cystic appearance alone. Ask whether the internal organization and boundary support the
↪  same interpretation or are unrelated, nonspecific findings. Here they provide mutually supporting
↪  positive evidence. The conclusion comes from coherence, not one observation acting as an automatic
↪  trigger.

**Decision**

The combined visible morphology provides coherent positive support, so `echinococcosis` is favored. This
↪  creates neither a fixed template nor a subtype requirement.

## Counterbalance against example bias

This example demonstrates the reasoning process, not a default preference for a positive diagnosis.

For every new image:

- Do not assume that a cystic or complex lesion is echinococcosis.
- Use only findings actually visible in that image.
- If the overall evidence is nonspecific or better supports a non-echinococcal lesion, classify it as
↪  `negative`.
- Do not force the new case to resemble the example.

## Output constraints

- Use only the existing fields: `image_valid`, `diagnosis`, `confidence`, `subtype`, and `findings`.
- Do not add fields.
- `diagnosis` must be exactly `echinococcosis`, `negative`, or `not_applicable`.
- For an invalid image, use `image_valid = false`, `diagnosis = not_applicable`, `subtype = null`, and
↪  `confidence = null`.
- Set `subtype = null` unless a subtype is very clearly supported.
- Keep `findings` concise and grounded in visible observations.
- Treat `confidence` as self-reported confidence, not as a calibrated probability.

## Provenance note

This experimental skill extends v3 with a worked reasoning example. The example was abstracted from
↪  source-provided morphology descriptions in the development data and rewritten as a general reasoning
↪  pattern. It does not include an evaluation image, image identifier, or verbatim source caption.

The source development image that inspired this example must not be counted in a later v4 development-set
↪  evaluation.
```